Bayesian Optimization Report: Function 8

GP Health Summary per Iteration

Iteration	GP Health	Status
1	0.666	Medium
2	0.672	Medium
3	0.668	Medium
4	0.671	Medium
5	0.671	Medium
6	0.674	Medium
7	0.680	Medium
8	0.676	Medium
9	0.679	Medium
10	0.681	Medium

GP Sigma Summary

Iteration	Mean Sigma	Min Sigma	Max Sigma
1	0.0825	0.0495	0.1210
2	0.0895	0.0419	0.1520
3	0.0857	0.0572	0.1221
4	0.0738	0.0536	0.1248
5	0.0811	0.0523	0.1216
6	0.0718	0.0477	0.1017
7	0.0816	0.0468	0.1537
8	0.0643	0.0462	0.0925
9	0.0665	0.0501	0.0894
10	0.0671	0.0455	0.0939

Best Results Per Iteration

Iteration	Best Input	Best Output	Kernel	Acquisition	GP Health	Anomalies
1	[0.02644871 0.25417413 0.03532238 0.06248106 0.90596252 0.63684762 0.23624164 0.52103246]	9.853830798070979	Matern(ν=0.5)	PORTFOLIO	0.666	None

2	[0.27932579 0.2427733 0.07357748 0.17099406 0.90631752 0.54586424 0.10292083 0.53738419]	9.880051388846612	Matern(ν=0.5)	PORTFOLIO	0.672	None
3	[0.02765367 0.1190652 0.12271456 0.22579316 0.83714701 0.56548265 0.25539036 0.47262727]	9.902666860514664	Matern(ν=0.5)	PORTFOLIO	0.668	None
4	[0.23501603 0.139237 0.19504023 0.10628002 0.83125323 0.64725708 0.13391826 0.40276932]	9.951605775633691	Matern(ν=0.5)	PORTFOLIO	0.671	None
5	[0.23591693 0.21350566 0.16276409 0.0976441 0.88578072 0.48944254 0.18504419 0.45671785]	9.99072502783371	Matern(ν=0.5)	PORTFOLIO	0.671	None
6	[0.17490731 0.28575384 0.17092112 0. 0.95631873 0.4493736 0.28233211 0.47138786]	9.955094836677771	Matern(ν=0.5)	PORTFOLIO	0.674	None
7	[0.3388386 0.16813638 0.2412462 0.1185063 0.93432484 0.54036507 0.21909137 0.40966935]	9.915488514306096	Matern(ν=0.5)	PORTFOLIO	0.680	None
8	[0.20881944 0.15438095 0.15642338 0.02455787 0.83813306 0.43154928 0.20568236 0.49280547]	9.988990173163073	Matern(ν=0.5)	PORTFOLIO	0.676	None
9	[0.22282438 0.26302324 0.16830463 0.15812394 0.87708227 0.40927914 0.28218942 0.38684571]	9.965492104897717	Matern(ν=0.5)	PORTFOLIO	0.679	None
10	[0.19597223 0.21821045 0.18539583 0.19946468 0.96934481 0.39656125 0.13182087 0.55650006]	9.980225852210838	Matern(ν=0.5)	PORTFOLIO	0.681	None

Acquisition Values Per Iteration

Iteration 1

Method	Max Value
UCB	6.840665
EI	0.021750
PI	0.349044
THOMPSON	6.789966
KG	0.021750
ENTROPY	36.955686

PORTFOLIO	4.691907

Iteration 2

Method	Max Value
UCB	6.787191
EI	0.021041
PI	0.312546
THOMPSON	6.846285
KG	0.021041
ENTROPY	36.932562
PORTFOLIO	4.830267

Iteration 3

Method	Max Value
UCB	6.753751
EI	0.026141
PI	0.373511
THOMPSON	6.826100
KG	0.026141
ENTROPY	36.543719
PORTFOLIO	4.690367

Iteration 4

Method	Max Value
UCB	6.762592
EI	0.018872
PI	0.338004
THOMPSON	6.794088
KG	0.018872
ENTROPY	36.114978
PORTFOLIO	4.673057

Iteration 5

Method	Max Value
UCB	6.777888

EI	0.022537
PI	0.352744
THOMPSON	6.846937
KG	0.022537
ENTROPY	36.637450
PORTFOLIO	4.678718

Iteration 6

Method	Max Value
UCB	6.803622
EI	0.021748
PI	0.376669
THOMPSON	6.888955
KG	0.021748
ENTROPY	36.579144
PORTFOLIO	4.659885

Iteration 7

Method	Max Value
UCB	6.858047
EI	0.020263
PI	0.325083
THOMPSON	6.947714
KG	0.020263
ENTROPY	37.815163
PORTFOLIO	4.760586

Iteration 8

Method	Max Value
UCB	6.808267
EI	0.014142
PI	0.327666
THOMPSON	6.860685
KG	0.014142
ENTROPY	36.307835

PORTFOLIO	4.650249

Iteration 9

Method	Max Value
UCB	6.845095
EI	0.016980
PI	0.339771
THOMPSON	6.874331
KG	0.016980
ENTROPY	36.421228
PORTFOLIO	4.674099

Iteration 10

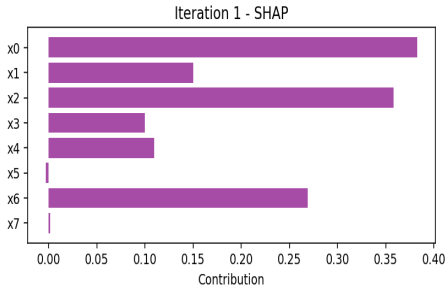
Method	Max Value
UCB	6.868808
EI	0.020940
PI	0.357375
THOMPSON	6.925970
KG	0.020940
ENTROPY	36.532215
PORTFOLIO	4.670573

LOO MSE Summary

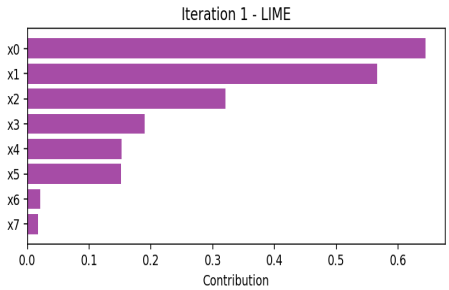
Iteration	LOO MSE
1	0.027403
2	0.025200
3	0.024845
4	0.026267
5	0.024700
6	0.024534
7	0.023450
8	0.019415
9	0.018897
10	0.018650

Feature Explanations (SHAP / LIME)

Iteration 1 - SHAP



Iteration 1 - LIME



Iteration 1 - SHAP Values

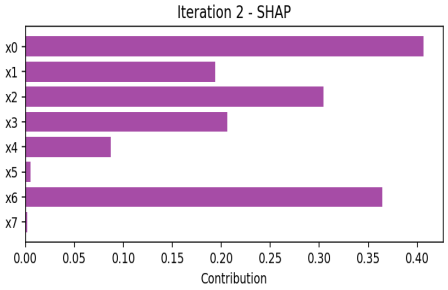
Feature	SHAP Value
x0	0.38253
x1	0.15020
x2	0.35797
x3	0.10022
x4	0.10992
x5	-0.00272
x6	0.26891
x7	0.00147

Iteration 1 - LIME Values

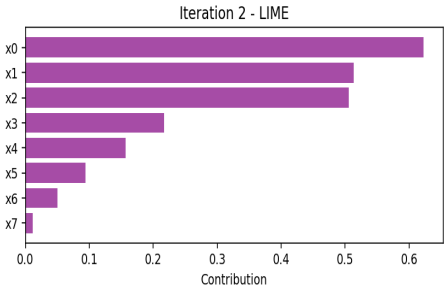
Feature	LIME Value
x0	0.64418
x1	0.56623
x2	0.32042
x3	0.19008
x4	0.15282

x5	0.15162
x6	0.02070
x7	0.01734

Iteration 2 - SHAP



Iteration 2 - LIME



Iteration 2 - SHAP Values

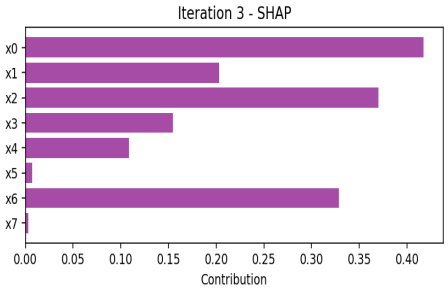
Feature	SHAP Value
x0	0.40622
x1	0.19375
x2	0.30427
x3	0.20634
x4	0.08710
x5	0.00504
x6	0.36421
x7	0.00204

Iteration 2 - LIME Values

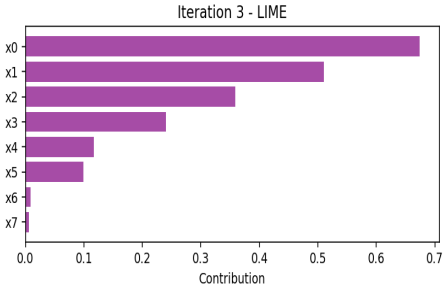
Feature	LIME Value
x0	0.62234
x1	0.51361

x2	0.50524
x3	0.21697
x4	0.15713
x5	0.09399
x6	0.05031
x7	0.01164

Iteration 3 - SHAP



Iteration 3 - LIME



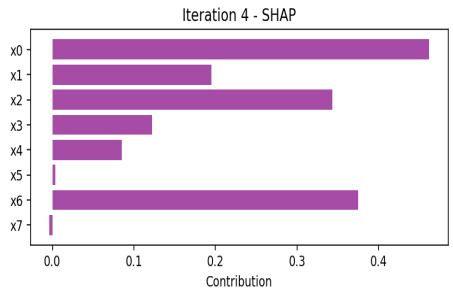
Iteration 3 - SHAP Values

Feature	SHAP Value
x0	0.41710
x1	0.20272
x2	0.36983
x3	0.15472
x4	0.10851
x5	0.00706
x6	0.32866
x7	0.00302

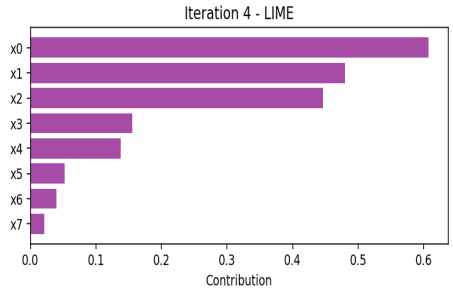
Iteration 3 - LIME Values

Feature	LIME Value
x0	0.67386
x1	0.51053
x2	0.35879
x3	0.23987
x4	0.11722
x5	0.09953
x6	0.00918
x7	0.00557

Iteration 4 - SHAP



Iteration 4 - LIME



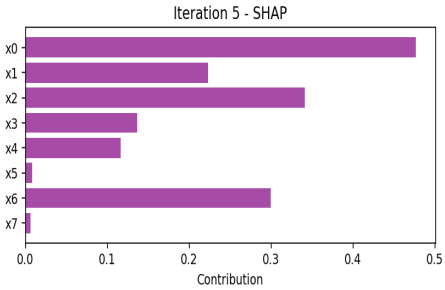
Iteration 4 - SHAP Values

Feature	SHAP Value
x0	0.46212
x1	0.19476
x2	0.34298
x3	0.12221
x4	0.08545
x5	0.00362
x6	0.37493
x7	-0.00360

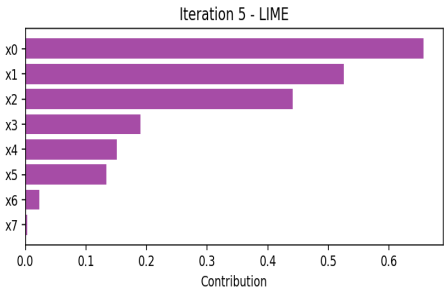
Iteration 4 - LIME Values

Feature	LIME Value
x0	0.60726
x1	0.47993
x2	0.44615
x3	0.15546
x4	0.13747
x5	0.05217
x6	0.03953
x7	0.02134

Iteration 5 - SHAP



Iteration 5 - LIME



Iteration 5 - SHAP Values

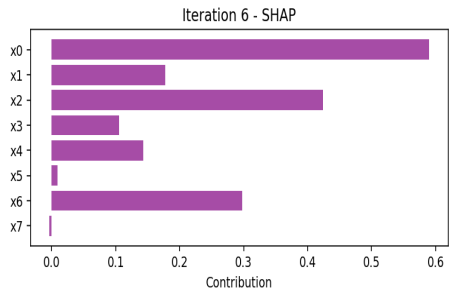
Feature	SHAP Value
x0	0.47713
x1	0.22329
x2	0.34157
x3	0.13639
x4	0.11606
x5	0.00818
x6	0.29939

x7	0.00600

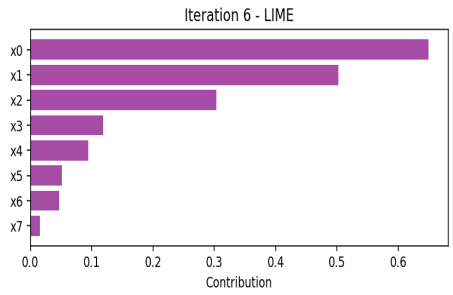
Iteration 5 - LIME Values

Feature	LIME Value
x0	0.65663
x1	0.52508
x2	0.44055
x3	0.18972
x4	0.15092
x5	0.13373
x6	0.02339
x7	0.00291

Iteration 6 - SHAP



Iteration 6 - LIME



Iteration 6 - SHAP Values

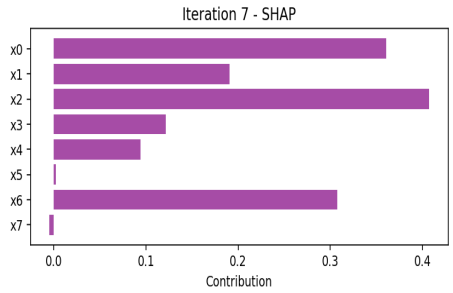
Feature	SHAP Value
x0	0.59016
x1	0.17760
x2	0.42394
x3	0.10526

x4	0.14272
x5	0.00894
x6	0.29745
x7	-0.00359

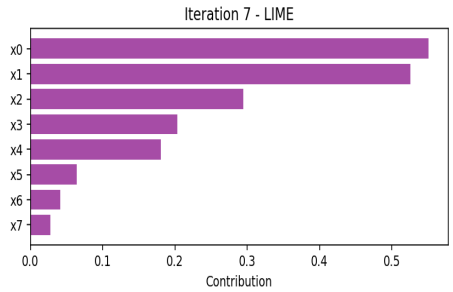
Iteration 6 - LIME Values

Feature	LIME Value
x0	0.64923
x1	0.50217
x2	0.30299
x3	0.11831
x4	0.09430
x5	0.05139
x6	0.04671
x7	0.01588

Iteration 7 - SHAP



Iteration 7 - LIME



Iteration 7 - SHAP Values

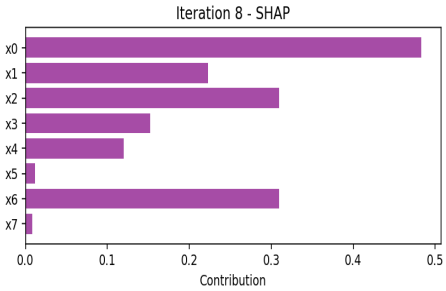
Feature	SHAP Value
x0	0.36065

x1	0.19060
x2	0.40758
x3	0.12135
x4	0.09411
x5	0.00216
x6	0.30787
x7	-0.00485

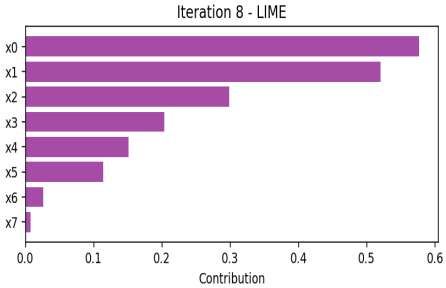
Iteration 7 - LIME Values

Feature	LIME Value
x0	0.55082
x1	0.52608
x2	0.29435
x3	0.20304
x4	0.18063
x5	0.06433
x6	0.04143
x7	0.02782

Iteration 8 - SHAP



Iteration 8 - LIME



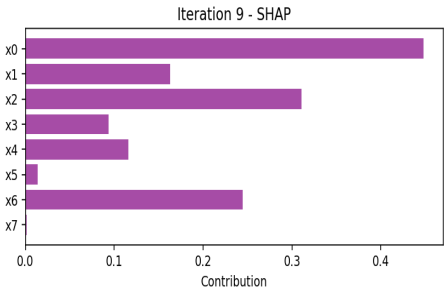
Iteration 8 - SHAP Values

Feature	SHAP Value
x0	0.48277
x1	0.22312
x2	0.30981
x3	0.15209
x4	0.12008
x5	0.01194
x6	0.30949
x7	0.00802

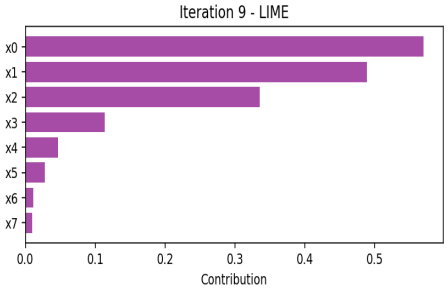
Iteration 8 - LIME Values

Feature	LIME Value
x0	0.57653
x1	0.52010
x2	0.29879
x3	0.20353
x4	0.15076
x5	0.11423
x6	0.02623
x7	0.00761

Iteration 9 - SHAP



Iteration 9 - LIME



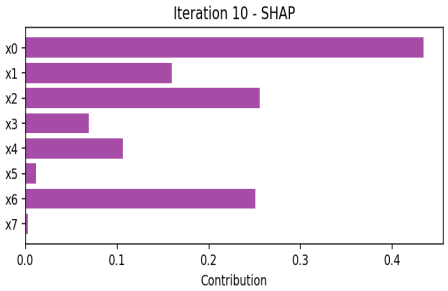
Iteration 9 - SHAP Values

Feature	SHAP Value
x0	0.44783
x1	0.16263
x2	0.31054
x3	0.09384
x4	0.11571
x5	0.01403
x6	0.24422
x7	0.00123

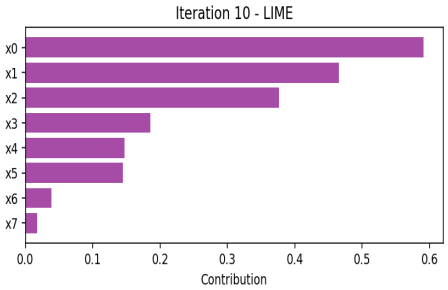
Iteration 9 - LIME Values

Feature	LIME Value
x0	0.56956
x1	0.48847
x2	0.33513
x3	0.11355
x4	0.04681
x5	0.02791
x6	0.01130
x7	0.00951

Iteration 10 - SHAP



Iteration 10 - LIME



Iteration 10 - SHAP Values

Feature	SHAP Value
x0	0.43377
x1	0.15971
x2	0.25513
x3	0.06894
x4	0.10651
x5	0.01145
x6	0.25045
x7	0.00256

Iteration 10 - LIME Values

Feature	LIME Value
x0	0.59085
x1	0.46536
x2	0.37607
x3	0.18570
x4	0.14705
x5	0.14464
x6	0.03847
x7	0.01746